

Tripling on Hessian curves via isogeny decomposition

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April 10, 2026

Abstract

We provide a new interpretation of the arithmetic on Hessian Kummer lines using level-3 theta structures. This allows us to break the record for tripling on elliptic curves and their Kummer lines, requiring only 4 multiplications and 4 squarings per tripling for well-chosen curve parameters.

1 Introduction

Elliptic curves have been a staple in public-key cryptography for decades, both for classical and post-quantum security. From a mathematical point of view, Weierstraß equations provide the most generality, but for cryptographic purposes, other models are often preferred due to their efficiency. Among this latter group, we have Edwards curves [3, 15], Huff curves [17, 12], Hessian curves, [6, 4], and even curves in Tate normal form [5, 8]. All of these models come with certain trade-offs, such as requiring points of particular (small) order.

One of the most celebrated discoveries is the Montgomery ladder on the Montgomery form of elliptic curves [2]. Not only is it extremely efficient, it also provides an easy constant-time implementation which is a highly sought-after property to combat side-channel attacks. Even in isogeny-based cryptography, Montgomery curves yield one of the most efficient representations to compute isogenies [18].

We revisit twisted Hessian curves and use an interpretation originating from level-3 theta structures, similar to the level-2 theta structures used in [19]. The basis of this analysis is Theorem 3.2 which described 3-isogenies of elliptic curves in Hesse form. This result can be viewed as a special case of the theory developed in [9]. This article provides a description of ℓ -isogenies of abelian varieties using level- ℓ theta structures with a focus on dimension two and $\ell = 3$. Here, we provide independent and elementary proofs for the special case of elliptic curves in Hessian form. Moreover, we extend the formulas to the case of twisted Hessian curves, as well as the corresponding Kummer lines.

With this new interpretation, we obtain a record for tripling on Kummer lines, namely a mere $4\mathbf{M} + 4\mathbf{S}$ for well-chosen parameters. The previous state-

of-the-art for tripling was $8\mathbf{M} + 4\mathbf{S}$ on a Hessian curve [1] (or $6\mathbf{M} + 6\mathbf{S}$ if multiplication with a cubic root of unity is cheap), and $4\mathbf{M} + 6\mathbf{S}$ on the Montgomery line [10].

Outline

In Section 2 we will give some background regarding Hessian curves, both untwisted and twisted versions; most of the material in this section is not new. In Section 3 we will discuss the interpretation of the level-3 theta structure on the Hessian curve. In Section 4 we will generalize these results to twisted Hessian curves and their Kummer lines. In particular, we provide an efficient tripling algorithm in that section.

Notation and conventions

We will work over fields k of characteristic not two nor three, with $\omega \in \bar{k}$ a fixed cubic root of unity. To avoid inversions, we will use projective coordinates throughout, with notation $(X : Y : Z) \in \mathbb{P}(k)^3$ and $(X : U) \in \mathbb{P}(k)^2$. For certain proofs, we may resort to affine coordinates and will denote those $(x, y) \in \mathbb{A}(k)^2$. The notations $\mathbf{M}, \mathbf{S}, \mathbf{M}_d$ and $\mathbf{A}$ will be used to respectively denote a multiplication, a squaring, a multiplication by a scalar d , and an addition. In general, additions are cheap compared to other arithmetic operations so they will be ignored for total operation counts. Isogenies projected on the Kummer line induce rational maps, but we will refer to them as isogenies as well since there is no potential for confusion.

Acknowledgments

Thomas Decru is in part funded by Fonds voor Wetenschappelijk Onderzoek (FWO) with reference number 1245025N, supported by the European Research Council (ERC) under the European Union’s Horizon 2020 research and innovation programme (grant agreement ISOCRYPT – No. 101020788), by the Research Council KU Leuven grant C14/24/099 and by CyberSecurity Research Flanders with reference number VOEWICS02. Sabrina Kunzweiler received funding from the French National Research Agency (ANR) under the ANR CIAO with reference ANR-19-CE48-0008, and the France 2030 program under grant ANR-22-PETQ-0008 (PQ-TLS).

2 Preliminaries

Hessian curves have been well-studied in the past; we refer to [1, 4] for some contemporary references. There is some ambiguity with regards to using a factor d or $3d$ in the defining equation of a Hessian curve. We opt for the latter since this makes our formulae most efficient, although we remark that for other applications the opposite may be true. In Subsection 2.1 we first cover the case

of untwisted Hessian curves. Subsection 2.2 reformulates the statements for twisted Hessian curves, where cubic twists are taken into account.

2.1 Hessian curves

Definition 2.1. A Hessian curve $\mathcal{H}_d/k$ is an elliptic curve defined by a polynomial of the form

$$X^3 + Y^3 + Z^3 = 3dXYZ,$$

where $d \in k$, $d^3 \neq 1$ and $(0 : -1 : 1)$ is the neutral element.

In Subsection 2.2, we will introduce twisted Hessian curves. To make the distinction clear, we sometimes refer to the standard Hessian curves as *untwisted* Hessian curves.

The j -invariant of $\mathcal{H}_d$ is given by

$$j(\mathcal{H}_d) = \left(3 \cdot \frac{d(8 + d^3)}{(d^3 - 1)} \right)^3,$$

see [11, Proposition 2].

It is well known that Hessian curves parametrize elliptic curves with full level-3 structure. More precisely, the 3-torsion of $\mathcal{H}_d$ is given by

$$\mathcal{H}_d[3] = \langle (-1 : 1 : 0), (0 : -\omega : 1), \rangle,$$

where $\omega \in \bar{k}$ is a primitive third root of unity. Moreover the action of 3-torsion points on $\mathcal{H}_d$ is described by the Heisenberg group of dimension 3. This is made explicit in the following lemma.

Lemma 2.2. Let $(X : Y : Z) \in \mathcal{H}_d$, and $P_1 = (-1 : 1 : 0)$, $P_2 = (0 : -\omega : 1)$. Then

$$(X : Y : Z) + P_1 = (Y : Z : X), \quad (X : Y : Z) + P_2 = (\omega^2 X : \omega Y : Z).$$

Proof. This follows from Mumford's theory on theta functions [16, Section 5]. An explicit description can be found in [11, Section 2.4]. $\square$

For any elliptic curve E with rational 3-torsion, there exists a rational transformation into a curve in Hessian form. The latter is determined by a choice of 3-torsion basis. Proposition 2.3 provides an explicit description of such a linear transformation.

Proposition 2.3. Let E be an elliptic curve in Weierstraß form defined over some field k and $E[3] = \langle P_1, P_2 \rangle$ with $P_1, P_2 \in E(k)$, and $\omega = e_3(P_1, P_2)$. Denote

$$P_1 = (x_1, y_1), \quad P_2 = (x_2, y_2), \quad P_1 + P_2 = (x_3, y_3), \quad \text{and } \delta = \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix},$$

Then the matrix $T = (t_{ij})_{i,j=1,2,3}$ with

$$\begin{aligned}
t_{11} &= -(x_1 - x_3) \cdot \delta \\
t_{12} &= 0 \\
t_{13} &= x_2(x_1 - x_3) \cdot \delta \\
t_{21} &= -t_{31} + (x_1 + (\omega - 1)x_2 - \omega x_3) \cdot \delta \\
t_{22} &= \omega(x_2 - x_3)(x_1 - x_3)(x_1 - x_2) \\
t_{23} &= -t_{33} - (\omega x_1 x_2 + (-\omega + 1)x_1 x_3 - x_2 x_3) \cdot \delta \\
t_{31} &= \omega(y_1 - y_3)(x_2 - x_3)(x_1 - x_2) \\
t_{32} &= -t_{22} \\
t_{33} &= -\omega(x_2 - x_3)(x_1 - x_2)(x_3 y_1 - x_1 y_3).
\end{aligned}$$

describes the coordinate transformation of E to a curve in Hessian form,

$$t : E \rightarrow \mathcal{H}_d, \quad P \mapsto T \cdot P.$$

Further, let $P_0 \in E \setminus E[3]$ and $(X_0 : Y_0 : Z_0) = t(P_0)$, then

$$d = \frac{X_0^3 + Y_0^3 + Z_0^3}{3X_0 Y_0 Z_0}.$$

Proof. Let E be as in the statement of the proposition. Since E has full k -rational 3-torsion, there exists an isomorphism to a curve in Hessian form, $t : E \rightarrow \mathcal{H}_d$ for some $d \in k$. Note that E and $\mathcal{H}_d$ are projectively equivalent, since they are both plane cubics with a rational flex point. This means that the isomorphism is described by some matrix $T \in PGL_3(k)$.

In order to explicitly determine the matrix T , we fix the images of the 3-torsion group $E[3]$ in $\mathcal{H}_d$ (it suffices to fix the images of $0, P_1, P_2, P_1 + P_2$).

$$\begin{aligned}
t((0 : 1 : 0)) &= (0 : -1 : 1), \\
t(P_1) &= (-1 : 0 : 1), \\
t(P_2) &= (0 : -\omega^2 : 1), \\
t(P_3) &= (-\omega : 1 : 0).
\end{aligned}$$

The matrix T is then obtained by basic linear algebra.

Finally, we can compute $d \in k$ by evaluating t at an arbitrary point $P_0 \in E \setminus E[3]$, i.e. computing $t(P_0) = (X_0 : Y_0 : Z_0)$. Since P_0 is not a 3-torsion point, we have that $X_0 Y_0 Z_0 \neq 0$, hence the value for d can be computed by evaluating the curve equation for $\mathcal{H}_d$ at $t(P_0)$. $\square$

We remark that the description of the transformation matrix in Proposition 2.3 is independent of the coefficients of the Weierstraß equation of E . In particular, the formulae can be applied to elliptic curves in different forms such as short Weierstraß form, Montgomery form or Legendre form.

Restricted to the corresponding Kummer lines, the coordinate transformation from Proposition 2.3 considerably simplifies. To state this result, let us first define a model for the Hessian Kummer line.

We recall that multiplication by $[-1]$ is given by permuting the coordinates Y and Z , i.e. $[-1](X : Y : Z) = (X : Z : Y)$. Hence the map

$$\pi : \mathcal{H}_d \rightarrow \mathbb{P}^1, \quad (X : Y : Z) \mapsto (X : Y + Z)$$

defines a projection onto the Kummer line of $\mathcal{H}_d$, and we say that $\mathcal{K}_d = \mathcal{H}_d / \langle \pm 1 \rangle = \pi(\mathcal{H}_d)$ is the (*untwisted*) Hessian Kummer line.

Corollary 2.4. *Let E be an elliptic curve in Weierstraß form defined over some field k and $E[3] = \langle P_1, P_2 \rangle$. Let x_1, x_2, x_3 be the x -coordinates of P_1, P_2 and $P_1 + P_2$, respectively. Then*

$$T = \begin{pmatrix} -(x_1 - x_3) & x_2(x_1 - x_3) \\ (x_1 + (\omega - 1)x_2 - \omega x_3) & -(\omega x_1 x_2 + (-\omega + 1)x_1 x_3 - x_2 x_3) \end{pmatrix}$$

describes the coordinate transformation of the Kummer line $E / \langle \pm 1 \rangle$ to the Hessian Kummer line $\mathcal{K}_d$,

$$t : E / \langle \pm 1 \rangle \rightarrow \mathcal{K}_d, \quad P \mapsto T \cdot P.$$

Proof. This follows from Proposition 2.3. $\square$

We end this subsection by formulating some isogeny statements for untwisted Hessian curves. For the proofs, we refer to their twisted versions in the next subsection since it suffices to compose those with a scaling isomorphism to adjust to the untwisted form.

Proposition 2.5. *The isogeny $\phi_1 : \mathcal{H}_d \rightarrow \mathcal{H}_{d_1}$ with $\ker \phi_1 = \langle (0 : -\omega : 1) \rangle$ is given by*

$$(X : Y : Z) \mapsto (\delta_1 XYZ : X^3 + \omega^2 Y^3 + \omega Z^3 : X^3 + \omega Y^3 + \omega^2 Z^3),$$

where $\delta_1^3 = d^3 - 1$ and $d_1 = d / \delta_1$.

The isogeny $\phi_2 : \mathcal{H}_d \rightarrow \mathcal{H}_{d_2}$ with $\ker \phi_2 = \langle (1 : -1 : 0) \rangle$ is given by

$$(X : Y : Z) \mapsto (\delta_2 XYZ : X^2 Z + XY^2 + YZ^2 : X^2 Y + XZ^2 + Y^2 Z),$$

where $\delta_2^3 = 9(d^2 + d + 1)$ and $d_2 = (d + 2) / \delta_2$.

Proposition 2.6. *The isogeny $\phi_3 : \mathcal{H}_d \rightarrow \mathcal{H}_{d_3}$ with $\ker \phi_3 = \langle (1 : -\omega^2 : 0) \rangle$ is given by*

$$(X : Y : Z) \mapsto (\delta_3 XYZ : \omega X^2 Z + \omega^2 XY^2 + YZ^2 : \omega X^2 Y + \omega^2 XZ^2 + Y^2 Z),$$

where $\delta_3^3 = 9(\omega^2 d^2 + \omega d + 1)$ and $d_3 = (\omega^2 d + 2\omega) / \delta_3$.

The isogeny $\phi_4 : \mathcal{H}_d \rightarrow \mathcal{H}_{d_4}$ with $\ker \phi_4 = \langle (1 : -\omega : 0) \rangle$ is given by

$$(X : Y : Z) \mapsto (\delta_4 XYZ : \omega^2 X^2 Z + \omega XY^2 + YZ^2 : \omega^2 X^2 Y + \omega XZ^2 + Y^2 Z),$$

where $\delta_4^3 = 9(\omega d^2 + \omega^2 d + 1)$ and $d_4 = (\omega d + 2\omega^2) / \delta_4$.

Porism 2.7. *Using the notation from Proposition 2.5 and 2.6, the isogenies $\hat{\phi}_i : \mathcal{H}_{d_i} \rightarrow \mathcal{H}_d$, with $i \in \{1, 2, 3, 4\}$, are defined by $\ker \hat{\phi}_i = \langle (0 : -\omega : 1) \rangle$.*

2.2 Twisted Hessian curves

Definition 2.8. A twisted Hessian curve $\mathcal{H}_{a,d}/k$ is an elliptic curve defined by the polynomial

$$aX^3 + Y^3 + Z^3 = 3dXYZ,$$

where $a, d \in k$, $a(d^3 - a) \neq 0$ and $(0 : -1 : 1)$ is the neutral element.

First, we note whenever a is a cube in k , a twisted Hessian curve is k -isomorphic to a curve in (untwisted) Hessian form:

Lemma 2.9. Let $\mathcal{H}_{a,d}$ over k be an elliptic curve in twisted Hessian form, and let $\alpha \in \bar{k}$ with $\alpha^3 = a$. Then $\mathcal{H}_{a,d}$ is isomorphic to the curve $\mathcal{H}_{d/\alpha}$ in (untwisted) Hessian form.

Proof. The isomorphism $t : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{d/\alpha}$ is given by $(X : Y : Z) \mapsto (\alpha X : Y : Z)$. $\square$

We could not find a formula for the j -invariant in the literature, but it can be easily deduced from the formula for the j -invariant in the untwisted setting (Subsection 2.1) and the preceding lemma.

Corollary 2.10. The j -invariant of $\mathcal{H}_{a,d}$ is given by

$$j(\mathcal{H}_{a,d}) = \frac{1}{a} \left(3 \cdot \frac{d(8a + d^3)}{d^3 - a} \right)^3.$$

The parameter a is the “twisting parameter” since it determines a (cubic) twist of some curve in Hessian form.

Remark 2.11. The case $d = 0$ is not excluded in Definition 2.8, resulting in a simplified defining equation of $aX^3 + Y^3 + Z^3 = 0$ for $\mathcal{H}_{a,0}$. This curve has j -invariant 0, hence it is isomorphic to the well-known curve $E_0 : Y^2Z = X^3 + Z^3$, which is supersingular over fields of characteristic $p \equiv 2 \pmod{3}$. The automorphism of degree three for $\mathcal{H}_{a,0}$ is given by $(X : Y : Z) \mapsto (\omega X : Y : Z)$.

Denoting $P_i = (X_i : Y_i : Z_i) \in \mathcal{H}_{a,d}$ for $i \in \{1, 2\}$, the arithmetic on $\mathcal{H}_{a,d}$ can be described by $-P_1 = (X_1 : Z_1 : Y_1)$ and $P_1 + P_2$ equaling

$$(X_1^2 Y_2 Z_2 - X_2^2 Y_1 Z_1 : Z_1^2 X_2 Y_2 - Z_2^2 X_1 Y_1 : Y_1^2 X_2 Z_2 - Y_2^2 X_1 Z_1),$$

unless this expressions happens to equal $(0 : 0 : 0)$; in that case $P_1 + P_2$ equals¹

$$(Z_2^2 X_1 Z_1 - Y_1^2 X_2 Y_2 : Y_2^2 Y_1 Z_1 - X_1^2 X_2 Z_2 : X_2^2 X_1 Y_1 - Z_1^2 Y_2 Z_2).$$

Since multiplication by -1 acts by permuting the variables Y and Z , a projection $\pi : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{a,d}/\langle \pm 1 \rangle$ is given by

$$\pi : (X : Y : Z) \mapsto (X : U) = (X : Y + Z).$$

¹Remark that at least one of them will be different from $(0 : 0 : 0)$ (see [1]), which makes them universally applicable. Generically, though, both of them work almost always..

We will call $\mathcal{H}_{a,d}/\langle \pm 1 \rangle$ the *twisted Hessian Kummer line* and denote it by $\mathcal{K}_{a,d}$.

The 3-torsion of a twisted Hessian curve is fully defined over $\bar{k}$ and explicit formulae for all outgoing 3-isogenies are known as well.

Proposition 2.12. *We have that*

$$\mathcal{H}_{a,d}[3] = \langle (0 : -\omega : 1), (1 : -\alpha : 0) \rangle,$$

with $\alpha^3 = a$, and consequently $(X : Y : Z) \in \mathcal{H}_{a,d}[3]$ iff $XYZ = 0$.

The isogeny $\phi_1 : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{a_1,d_1}$ with $\ker \phi_1 = \langle (0 : -\omega : 1) \rangle$ is given by

$$(X : Y : Z) \mapsto (XYZ : aX^3 + \omega^2Y^3 + \omega Z^3 : aX^3 + \omega Y^3 + \omega^2Z^3),$$

where $a_1 = d^3 - a$ and $d_1 = d$.

The isogeny $\phi_2 : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{a_2,d_2}$ with $\ker \phi_2 = \langle (1 : -\alpha : 0) \rangle$ is given by

$$(X : Y : Z) \mapsto (XYZ : \alpha^2X^2Z + \alpha XY^2 + YZ^2 : \alpha^2X^2Y + \alpha XZ^2 + Y^2Z),$$

where $a_2 = 9(\alpha d^2 + \alpha^2d + a)$ and $d_2 = d + 2\alpha$.

Proof. See [4, Theorem 1, Theorem 3]. □

The following formulae can be found, for example, in [4, Theorem 3] as well as [13]. We adapt the proof to our notation for the sake of completeness.

Proposition 2.13. *The isogeny $\phi_3 : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{a_3,d_3}$ with $\ker \phi_3 = \langle (1 : -\omega^2\alpha : 0) \rangle$ is given by*

$$(X : Y : Z) \mapsto (XYZ : \omega\alpha^2X^2Z + \omega^2\alpha XY^2 + YZ^2 : \omega\alpha^2X^2Y + \omega^2\alpha XZ^2 + Y^2Z),$$

where $a_3 = 9(\omega^2\alpha d^2 + \omega\alpha^2d + a)$ and $d_3 = \omega^2d + 2\omega\alpha$.

The isogeny $\phi_4 : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{a_4,d_4}$ with $\ker \phi_4 = \langle (1 : -\omega\alpha : 0) \rangle$ is given by

$$(X : Y : Z) \mapsto (XYZ : \omega^2\alpha^2X^2Z + \omega\alpha XY^2 + YZ^2 : \omega^2\alpha^2X^2Y + \omega\alpha XZ^2 + Y^2Z),$$

where $a_4 = 9(\omega\alpha d^2 + \omega^2\alpha^2d + a)$ and $d_4 = \omega d + 2\omega^2\alpha$.

Proof. This is essentially the same proof as [4, Theorem 3]. Focusing on ϕ_3 , write $F = XYZ$, $G = \omega\alpha^2X^2Z + \omega^2\alpha XY^2 + YZ^2$ and $H = \omega\alpha^2X^2Y + \omega^2\alpha XZ^2 + Y^2Z$ for the coordinate maps. One can verify that $a_3F^3 + G^3 + H^3 - 3d_3FGH$ has a factor $aX^3 + Y^3 + Z^3 - 3dXYZ$, such that $\text{Im}(\phi_3) \subseteq \mathcal{H}_{a_3,d_3}$. By irreducibility of elliptic curves, one must have $\text{Im}(\phi_3) = \mathcal{H}_{a_3,d_3}$. A straightforward computation shows that $\phi_3(\langle (-\omega : 1 : 0) \rangle) = (0 : -1 : 1)$ and $\phi_3(\mathcal{H}_{a,d}[3]) = \langle (0 : -\omega : 1) \rangle$. An analogous argument holds for ϕ_4 . □

All ϕ_i map the 3-torsion of $\mathcal{H}_{a,d}$ to a subgroup of the same form - albeit on different codomain curves $\mathcal{H}_{a_i,d_i}$ - which is not surprising given that they all have XYZ as first coordinate map. The following porism will thus allow us to easily split multiplication-by-3 on $\mathcal{H}_{a,d}$ in four distinct ways, namely as $\widehat{\phi}_i \circ \phi_i$, where $\widehat{\phi}_i$ denotes the dual isogeny of ϕ_i .

Porism 2.14. *Using the notation from Proposition 2.12 and 2.13, the isogenies $\hat{\phi}_i : \mathcal{H}_{a_i, d_i} \rightarrow \mathcal{H}_{a, d}$, with $i \in \{1, 2, 3, 4\}$, are defined by $\ker \hat{\phi}_i = \langle (0 : -\omega : 1) \rangle$.*

We end this subsection with defining a specific type of Hessian curve that will allow us the most efficient formulae for tripling.

Definition 2.15. *We say that $\mathcal{H}_{a, d}$ is in twisted normal form if $d = 1$ and will denote this curve by $\mathcal{H}_a$.*

Note that any curve in (twisted or untwisted) Hessian form admits an equation in twisted normal form.

Lemma 2.16. *Let $\mathcal{H}_{a, d}$ over k be an elliptic curve in twisted Hessian form. Then $\mathcal{H}_{a, d}$ is isomorphic to the curve $\mathcal{H}_{a/d^3}$ in twisted normal form.*

Proof. The isomorphism $t : \mathcal{H}_{a, d} \rightarrow \mathcal{H}_{a/d^3}$ is given by $(X : Y : Z) \mapsto (dX : Y : Z)$. $\square$

Remark 2.17. *The isomorphisms described in Lemmata 2.9 and 2.16 are both of the form*

$$t_\lambda : \mathcal{H}_{a, d} \rightarrow \mathcal{H}_{\lambda^3 a, \lambda d}, \quad (X : Y : Z) \mapsto (X : \lambda Y : \lambda Z).$$

for some $\lambda \in \bar{k}$. These isomorphisms allow us to translate results obtained on one curve forms, to results on the other curve forms.

3 3-isogenies by Fourier transforms

In this section we provide an alternative way to consider and compute 3-isogenies on Hessian curves, akin to the work of Robert on the computation of 2-isogenies [19] which has led to several follow-up works on the efficient computations of doublings, and isogeny chains in various dimensions [20, 7, 14]. We first explain the computation of isogenies on the Hessian curve itself, and then provide the induced maps on the Kummer line.

3.1 3-isogenies on Hessian curves

As we explained in Subsection 2.1, the action of the 3-torsion points is described by the Heisenberg group of dimension 3. This is similar to the setting of level-2 theta functions in [19] where the action of 2-torsion points is described by the Heisenberg group of dimension 2. In that setting, a 2-isogeny may be decomposed into three operations: coordinate-wise squaring, a Hadamard transformation, and coordinate-wise scaling. In the following, we show that an analogous decomposition can be achieved for 3-isogenies in Hessian form.

Notation 3.1. *For points $(X : Y : Z) \in \mathcal{H}_d$, we introduce the following operations.*

- C : *Coordinate-wise cubing; i.e. $C(X : Y : Z) = (X^3 : Y^3 : Z^3)$.*

- M : Discrete Fourier transform; i.e.

$$M(X : Y : Z) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}.$$

- $\mu_{x,u}$: Scaling; i.e. $\mu_{x,u}(X : Y : Z) = (xX : uY : uZ)$.

With this notation, we provide decompositions of the different 3-isogenies with domain $\mathcal{H}_d$. The most important building block will be the 3-isogeny ϕ_1 with kernel $\langle (0 : -\omega : 1) \rangle$.

Theorem 3.2. *Let $\mathcal{H}_d$ be an elliptic curve in Hessian form and d' such that $d' = d/\delta$, where $\delta^3 = d^3 - 1$. Then*

$$\phi_1 : (X : Y : Z) \mapsto \mu_{1,d'} \circ M \circ C(X : Y : Z)$$

defines an isogeny $\phi_1 : \mathcal{H}_d \rightarrow \mathcal{H}_{d'}$ with kernel $\ker(\phi_1) = \langle (0 : -\omega : 1) \rangle$.

Proof. This follows from comparing with existing formulae (for example Proposition 2.5). $\square$

Note that $\phi_1(\mathcal{H}_d[3]) = \langle (0 : -\omega : 1) \rangle \subset \mathcal{H}_{d'}$, hence the dual isogeny $\hat{\phi}_1$ can be described in the same way as ϕ_1 , and we obtain an easy description of multiplication by [3] on $\mathcal{H}_d$. We remark that this representation is similar to the formula in [1, Section 6, third example], where the authors also observe that the discrete Fourier transform plays a role in the computation of triplings. In particular, disregarding multiplication by constants, both our formula and the formula in [1] have the same cost.

Corollary 3.3. *Let $\mathcal{H}_d$ be an elliptic curve in Hessian form and d' such that $d' = d/\delta$, where $\delta^3 = d^3 - 1$. Then the multiplication-by-3 map on $\mathcal{H}_d$ can be evaluated as*

$$[3] : (X : Y : Z) \mapsto \mu_{1,d} \circ M \circ C \circ \mu_{1,d'} \circ M \circ C(X : Y : Z).$$

In particular, it can be evaluated in $6\mathbf{M} + 6\mathbf{S} + 1\mathbf{M}_{1/d} + 1\mathbf{M}_{1/d'}$ (in a field where the cost for multiplication by ω is negligible).

Proof. The formula is an immediate consequence of Theorem 3.2. For the operation count we note that C is evaluated using 3 cubings, i.e. $3(\mathbf{M} + \mathbf{S})$, and $\mu_{1,d}$ is evaluated using one scalar multiplication by $1/d$ (similarly for $\mu_{1,d'}$). The evaluation of the discrete Fourier transform (M) only requires multiplications by ω . $\square$

In order to describe the remaining 3-isogenies, we may simply transform the kernel into the right form. For this purpose, consider $P = (-\epsilon : 1 : 0) \in \mathcal{H}_d[3] \setminus \langle (0 : -\omega : 1) \rangle$, hence $\epsilon^3 = 1$. First, observe that after a scaling isomorphism $\mu_{1,\epsilon} : \mathcal{H}_d \rightarrow \mathcal{H}_{\epsilon d}$, we may assume that $P = (-1 : 1 : 0)$. Transforming the latter into a kernel generator of the right form is achieved by another application of the Fourier transform.

Lemma 3.4. *Let $\mathcal{H}_d$ be a curve in Hessian form and denote $P_2 = (-1 : 1 : 0) \in \mathcal{H}_d[3]$. We set*

$$\tilde{d} = \frac{d+2}{d-1}.$$

Then the discrete Fourier transform defines an isomorphism

$$M : \mathcal{H}_d \rightarrow \mathcal{H}_{\tilde{d}}, \quad P \mapsto M \cdot P$$

and $M(P_2) = \tilde{P}_1 = (0 : -\omega : 1) \in \mathcal{H}_{\tilde{d}}$.

As a consequence of Theorem 3.2 and the preceding lemma, we obtain the following description of (the remaining) 3-isogenies on the Hessian curves.

Corollary 3.5. *Let $\mathcal{H}_d$ be an elliptic curve in Hessian form with $P = (-\epsilon : 1 : 0)$ a 3-torsion point on $\mathcal{H}_d$, where $\epsilon^3 = 1$. We set*

$$\tilde{d} = \frac{\epsilon d + 2}{\epsilon d - 1}, \quad \tilde{\delta} = \sqrt[3]{\tilde{d}^3 - 1}, \quad d' = \frac{\tilde{d}}{\tilde{\delta}}.$$

Then

$$\phi : (X : Y : Z) \mapsto \mu_{1,d'} \circ M \circ C(X : Y : Z) \circ M \circ \mu_{1,\epsilon}$$

defines an isogeny $\phi_1 : \mathcal{H}_d \rightarrow \mathcal{H}_{d'}$ with kernel $\langle (0 : -\omega : 1) \rangle$.

We note that the evaluation cost of the isogeny with kernel $\langle (0 : -\omega : 1) \rangle$ (Theorem 3.2) and the remaining 3-isogenies (Corollary 3.5) only differs by additions and multiplications by a third root of unity ω .

3.2 3-isogenies on Hessian Kummer lines

The formulas derived in the previous subsection can be easily adapted to formulas on the Hessian Kummer line $\mathcal{K}_d$. One can readily verify that the elementary operations C (cubing), M (discrete Fourier transform) and $\mu_{x,u}$ (scaling) push down on the Hessian Kummer line $\mathcal{K}_d$ as follows, where we slightly abuse notation and keep the names.

- $M : (X : U) \mapsto (X + U : 2X - U)$
- $C_d : (X : U) \mapsto (X^2(U + dX) : U(dU^2 - X^2))$
- $\mu_{x,u} : (X : U) \mapsto (xX : uU)$.

Note that the evaluation of the cubing in terms of Kummer coordinates depends on the curve coefficient d , therefore we use the notation C_d .

This description allows us to translate the formulas for the 3-isogenies of Hessian curves from Subsection 3.1 to easy-to-evaluate maps on the corresponding Kummer lines.

Proposition 3.6. *We use the notation from Theorem 3.2 and Corollary 3.5, and consider the isogenies $\phi_i : \mathcal{H}_d \rightarrow \mathcal{H}_{d_i}$, with kernel generated by P_i for $i \in \{1, 2\}$, where*

$$P_1 = (0 : -\omega : 1), P_2 = (-\epsilon : 1 : 0)$$

for some ϵ with $\epsilon^3 = 1$. These isogenies induce maps $\varphi_i : \mathcal{K}_d \rightarrow \mathcal{K}_{d_i}$ on the corresponding Kummer lines and can be evaluated in $2\mathbf{M} + 2\mathbf{S} + 3\mathbf{M}_d$.

Explicitly, it holds that

$$\varphi_1(X : U) = \mu_{1,d_1} \circ M \circ C_d(X : U)$$

and

$$\varphi_2(X : U) = \mu_{1,d_2} \circ M \circ C_d \circ M \circ \mu_{1,\epsilon}(X : U).$$

Proof. This follows from Theorem 3.2, Corollary 3.5 and the description of the maps $C, M, \mu_{x,u}$ on the Kummer lines.

For the operation count, we note that the cubing on the Kummer surface requires 2 squaring and 2 multiplications, as well as 2 scalar multiplications by d (respectively $\tilde{d}$). The scaling map only requires one scalar multiplication by the scalar d_i depending on d . $\square$

Lemma 3.7. *Let $\mathcal{H}_d$ be a Hessian curve. Then $dX + Y + Z = 0$ iff $(X : Y : Z) = (0 : -1 : 1)$.*

Proof. Substituting $Z = -(dX + Y)$ in the defining equation of $\mathcal{H}_d$, we find

$$X^3 + Y^3 - (dX + Y)^3 = -3dXY(dX + Y),$$

which simplifies to $(d^3 - 1)X^3 = 0$. Since $d^3 \neq 1$, we must have that $X = 0$, which only holds for $(X : Y : Z) \in \langle (0 : -\omega : 1) \rangle$ by Proposition 2.5, and the result follows immediately. $\square$

For completeness, we also provide concrete formulas for the 3-isogenies. These formulas may either be deduced by symbolically evaluating the description of the 3-isogenies in Proposition 3.6, or they can be deduced from Propositions 2.5 and 2.6.

Proposition 3.8. *Using the notation from Proposition 2.5 and 2.6, the isogenies $\phi_i : \mathcal{H}_d \rightarrow \mathcal{H}_{d_i}$, with $i \in \{1, 2, 3, 4\}$ induce isogenies $\varphi_i : \mathcal{K}_d \rightarrow \mathcal{K}_{d_i}$ which are given by*

$$\begin{aligned} \varphi_1(X : U) &\mapsto (\delta_1(X^3 + U^3) : d(2X^3 - U^3) + 3X^2U) \\ \varphi_2(X : U) &\mapsto (3d\delta_2X(X^2 - UX + U^2) : -2X^3 + 3(1 + d)UX^2 + U^3), \\ \varphi_3(X : U) &\mapsto (3d\delta_3X(\omega X^2 - \omega^2UX + U^2) : -2X^3 + 3\omega^2(\omega^2 + d)UX^2 + U^3), \\ \varphi_4(X : U) &\mapsto (3d\delta_4X(\omega^2X^2 - \omega UX + U^2) : -2X^3 + 3\omega(\omega + d)UX^2 + U^3). \end{aligned}$$

Proof. The formulae are obtained by algebraic manipulations of the isogeny formulae from Propositions 2.5 and 2.6. To illustrate this proof strategy, we outline the computations for the first isogeny.

By Lemma 3.7, we can assume that $dX + Y + Z \neq 0$. Let $U = Y + Z$ and note that

$$YZ = \frac{X^3 + U^3}{3(dX + U)}$$

holds. As per Proposition 2.5, the isogeny ϕ_1 is given by $\phi_1 : (X : Y : Z) \mapsto (X_1 : Y_1 : Z_1)$, with

$$(X_1 : Y_1 : Z_1) = (XYZ : X^3 + \omega^2 Y^3 + \omega Z^3 : X^3 + \omega Y^3 + \omega^2 Z^3).$$

This implies that $\varphi_1 : (X : U) \mapsto (X_1 : U_1)$, where

$$\begin{aligned} (X_1 : U_1) &= (XYZ : (X^3 + \omega^2 Y^3 + \omega Z^3) + (X^3 + \omega Y^3 + \omega^2 Z^3)) \\ &= \left(X \frac{X^3 + U^3}{3(dX + U)} : 2X^3 - (Y^3 + Z^3) \right) \\ &= (X(X^3 + U^3) : (2X^3 - U^3)3(dX + U) + 3U(X^3 + U^3)) \\ &= (X^3 + U^3 : 3d(2X^3 - U^3) + 9X^2U). \end{aligned}$$

The formulae for the remaining isogenies can be obtained by similar computations. $\square$

4 Decomposition of 3-isogenies in the twisted setting

Similar as in the previous section, we can translate the 3-isogenies with domain a twisted Hessian curve $\mathcal{H}_{a,d}$ to isogenies with domain the twisted Hessian Kummer line $\mathcal{K}_{a,d}$.

Proposition 4.1. *Using the notation from Proposition 2.12 and 2.13, the isogenies $\phi_i : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{a_i,d_i}$, with $i \in \{1, 2, 3, 4\}$ induce isogenies $\varphi_i : \mathcal{K}_{a,d} \rightarrow \mathcal{K}_{a_i,d_i}$ which are given by*

$$\begin{aligned} \varphi_1 : (X : U) &\mapsto (aX^3 + U^3 : d(2aX^3 - U^3) + 9aX^2U) \\ \varphi_2 : (X : U) &\mapsto (X(\alpha^2 X^2 - \alpha XU + U^2) : -2aX^3 + 3\alpha(\alpha + d)X^2U + U^3), \\ \varphi_3 : (X : U) &\mapsto (X(\omega\alpha^2 X^2 - \omega^2\alpha XU + U^2) : -2aX^3 + 3\omega^2\alpha(\omega^2\alpha + d)X^2U + U^3), \\ \varphi_4 : (X : U) &\mapsto (X(\omega^2\alpha^2 X^2 - \omega\alpha XU + U^2) : -2aX^3 + 3\omega\alpha(\omega\alpha + d)X^2U + U^3). \end{aligned}$$

Proof. The formulae are obtained by algebraic manipulations of the isogeny formulae from Propositions 2.12 and 2.13. The proof is analogous to the proof of Proposition 3.8 and we omit it here. $\square$

4.1 3-isogenies on twisted Hessian curves

Again, we want to decompose the isogeny evaluation into different steps for efficiency purposes. Before describing the decomposition on twisted Kummer lines, we briefly explain the ideas on twisted Hessian curves.

We restrict our attention to the isogeny ϕ_1 with kernel $\ker(\phi_1) = \langle (0 : -\omega : 1) \rangle$ as this is the main building block, and all other 3-isogenies can be obtained by composition with a discrete Fourier transform (cf. Subsection 3.2). We slightly modify the cubing operation to account for the twisting parameter a . The necessity of this modification will become clear in the proof of Proposition 4.2.

$$\bullet C_{a,d} : (X : Y : Z) \mapsto (aX^3 : Y^3 : Z^3)$$

Proposition 4.2. *Let $\mathcal{H}_{a,d}$ be an elliptic curve in twisted Hessian form. Let $d' = d$ and $a' = d^3 - a$. Then the isogeny $\phi_1 : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{a',d'}$ is given by*

$$\phi_1(X : Y : Z) = \mu_{1,d} \circ M \circ C_a(X : Y : Z)$$

and can be evaluated in $6\mathbf{M} + 6\mathbf{S} + 1\mathbf{M}_a + 2\mathbf{M}_d$.

Proof. The main idea is to first apply a transformation into (untwisted) Hessian form using Lemma 2.9, and then apply our main result, Theorem 3.2.

The transformation in untwisted Hessian form is given by $t : (X : Y : Z) \mapsto (\alpha X : Y : Z)$. This corresponds to an isomorphism $\mathcal{H}_{a,d} \rightarrow \mathcal{H}_{d/\alpha}$. Now on the latter curve, the 3-isogeny ϕ_1 is given by

$$(X : Y : Z) \mapsto \mu_{1,d_1} \circ M \circ C(X : Y : Z), \quad \text{where } d_1 = \frac{d/\alpha}{\sqrt[3]{d^3/a - 1}}.$$

First note that composing t with the cubing C , we obtain

$$C \circ t(X : Y : Z) = (aX^3 : Y^3 : Z^3) = C_a.$$

This explains the change in the description of the “cubing” map.

Note that working on twisted Hessian curves, it is technically not necessary to apply the scaling map μ_{1,d_1} , since we do not require $a = 1$. We note that the inverse of the scaling map is

$$\mu_{1,d_1}^{-1} = \mu_{1,1/d_1} : \mathcal{H}_{d_1} \rightarrow \mathcal{H}_{1/d_1^3,1} = \mathcal{H}_{(d^3-a)/d^3,1}.$$

This means that we have an isogeny

$$\mathcal{H}_{a,d} \rightarrow \mathcal{H}_{(d^3-a)/d^3,1}, \quad (X : Y : Z) \mapsto M \circ C_a(X : Y : Z).$$

In order to avoid an inversion when computing the codomain value for a , we compose this with the scaling $\mu_{1,d} : \mathcal{H}_{(d^3-a)/d^3,1} \rightarrow \mathcal{H}_{d^3-a,d}$. This proves the isogeny formula provided in the Proposition. $\square$

Remark 4.3. *For some applications (such as multiplication by 3, where one works on a fixed elliptic curve), it can be preferable to work with the isogeny*

$$\phi_1' : \mathcal{H}_{a,d} \rightarrow \mathcal{H}_{(d^3-a)/d^3,1}, \quad (X : Y : Z) \mapsto M \circ C_a(X : Y : Z)$$

that appeared as an intermediate step in the proof of Proposition 4.2. This saves two scalar multiplication at each evaluation, at the cost of computing one inversion during the computation of the codomain parameters.

4.2 3-isogenies on twisted Hessian Kummer lines

To translate the results from twisted Hessian curves to twisted Kummer lines, we proceed as in Subsection 3.2. That is, we translate the elementary operations *discrete Fourier transform, scaling and cubing* to the Kummer line. The maps M and $\mu_{x,u}$ are defined in Subsection 3.2:

- $M : (X : U) \mapsto (X + U : 2X - U)$
- $\mu_{x,u} : (X : U) \mapsto (xX : uU)$.

For the cubing map, we C_a , we obtain:

- $C_{a,d} : (X : U) \mapsto (aX^2(U + dX) : U(dU^2 - aX^2))$

Proposition 4.4. *The isogeny φ_1 from Proposition 3.8 can be evaluated in $2\mathbf{M} + 2\mathbf{S} + 3\mathbf{M}_d + 1\mathbf{M}_a$.*

Explicitly, it holds that

$$\varphi_1(X : U) = \mu_{1,d} \circ M \circ C_{a,d}(X : U).$$

Proof. This is a consequence of Proposition 4.2. $\square$

We can decompose multiplication-by-3 on $\mathcal{H}_{a,d}$ as $\hat{\phi}_i \circ \phi_i$ for any $i \in \{1, 2, 3, 4\}$. Using the formula for ϕ_1 from Proposition 4.4 this yields particularly efficient multiplication-by-3 formulas when $d = 1$, i.e. when $\mathcal{H}_{a,d} = \mathcal{H}_a$ is in twisted normal form (cf. Definition 2.15).

Corollary 4.5. *On any twisted Hessian Kummer line $\mathcal{K}_{a,d}$, multiplication-by-3 can be evaluated in $4\mathbf{M} + 4\mathbf{S} + 6\mathbf{M}_d + 2\mathbf{M}_a$. Moreover, if the equation is in twisted normal form (Definition 2.15), multiplication-by-3 can be evaluated in $4\mathbf{M} + 4\mathbf{S} + 2\mathbf{M}_a$.*

Proof. This is applying the formula for φ_1 from Proposition 3.8 twice. The algorithm can be found in Algorithm 1. $\square$

Algorithm 1 Tripling on the Kummer line $\mathcal{K}_a$ in twisted Hessian normal form

Input: $P = (X : U)$ on $\mathcal{K}_a$

Output: $3 \cdot P$

- | | |
|--|--|
| 1: $S, T \leftarrow a \cdot X^2, U^2$ | $\triangleright 1\mathbf{M}_a + 2\mathbf{S}$ |
| 2: $X_1, U_1 \leftarrow S \cdot (U + X), U \cdot (T - S)$ | $\triangleright 2\mathbf{M}$ |
| 3: $X_2, U_2 \leftarrow X_1 + U_1, 2X_1 - U_1$ | |
| 4: $S', T' \leftarrow (1 - a) \cdot X_2^2, U_2^2$ | $\triangleright 1\mathbf{M}_a + 2\mathbf{S}$ |
| 5: $X_3, U_3 \leftarrow S' \cdot (U_2 + X_2), U_2 \cdot (T' - S')$ | $\triangleright 2\mathbf{M}$ |
| 6: $X_4, U_4 \leftarrow X_3 + U_3, 2X_3 - U_3$ | |
| 7: return $(X_4 : U_4)$ | $\triangleright$ Total Cost: $4\mathbf{M} + 4\mathbf{S} + 2\mathbf{M}_a$ |
-

Remark that this is significantly cheaper than the state-of-the-art tripling formulae for Kummer lines, which cost $4\mathbf{M} + 6\mathbf{S}$ on the Montgomery line [10,

Algorithm 46]. As a comparison, we also provide the most efficient doubling formulae to show that tripling on the Kummer line in twisted Hessian normal form is almost as cheap.

Proposition 4.6. *Let $P = (X : U) \in \mathcal{K}_{a,d}$, then*

$$2 \cdot P = (X \cdot (-aX^3 + U^2 \cdot (3dX + 2U)) : -aX^3(3dX + 4U) - U^4).$$

Proof. The formulae are deduced from the addition formula in [1, Theorem 4.2] on the curve $\mathcal{H}_{a,d}$. $\square$

Corollary 4.7. *Doubling a point on $\mathcal{K}_{a,d}$ can be evaluated in $4\mathbf{M} + 3\mathbf{S} + 1\mathbf{M}_a + 1\mathbf{M}_d$.*

Proof. This follows from Proposition 4.6. Pseudo-code for the efficient evaluation of the formula is provided in Algorithm 2. $\square$

Algorithm 2 Doubling on the twisted Kummer line $\mathcal{K}_{a,d}$

Input: $P = (X : U)$ on $\mathcal{K}_{a,d}$

Output: $2 \cdot P$

1: $X_2, X_3 \leftarrow X^2, X^3$	$\triangleright 1\mathbf{M} + 1\mathbf{S}$
2: $U_2, U_4 \leftarrow U^2, U^4$	$\triangleright 2\mathbf{S}$
3: $S \leftarrow a \cdot X_3$	$\triangleright 1\mathbf{M}_a$
4: $T \leftarrow d \cdot X$	$\triangleright 1\mathbf{M}_d$
5: $\tilde{X} \leftarrow X \cdot (-S + U_2 \cdot (3T + 2U))$	$\triangleright 2\mathbf{M}$
6: $\tilde{U} \leftarrow -S \cdot (3T + 4U) - U_4$	$\triangleright 1\mathbf{M}$
7: return $(\tilde{X} : \tilde{U})$	$\triangleright$ Total Cost: $4\mathbf{M} + 3\mathbf{S} + 1\mathbf{M}_a + 1\mathbf{M}_d$

5 Conclusion

We provided an interpretation of the level-3 theta structure of Hessian curves to obtain state-of-the-art tripling formulae on Hessian Kummer lines in twisted normal form. They require a mere $4\mathbf{M} + 4\mathbf{S}$ for well-chosen parameters.

A natural question is whether this can be extended to build a ladder similar to the double-and-add in the Montgomery form, but then a triple-and-add-or-subtract. Unfortunately for that we require somewhat efficient differential addition as well, since one cannot distinguish between P and $-P$ on the Kummer line. To the best of our knowledge, no such efficient differential addition formulae exist. We provide the best we could find in Algorithm 3, though it would need a massive improvement to be able to compete with the Montgomery ladder. We leave further exploration for future work.

Algorithm 3 Differential addition on the twisted Kummer line $\mathcal{K}_{a,d}$

Input: $P = (X_1 : U_1)$, $Q = (X_2 : U_2)$ and $P - Q = (X_3 : U_3) \neq (0 : 0)$ on $\mathcal{K}_{a,d}$

Output: $P + Q$

```
1:  $R_1, R_2, R_3, R_4 \leftarrow X_1 \cdot X_2, X_1 \cdot U_2, X_2 \cdot U_1, U_1 \cdot U_2,$   $\triangleright 4\mathbf{M}$ 
2:  $S_1 \leftarrow a \cdot R_1$   $\triangleright 1\mathbf{M}_a$ 
3:  $T_1, T_2, T_3, T_4 \leftarrow d \cdot R_1, d \cdot R_2, d \cdot R_3, d \cdot R_4$   $\triangleright 4\mathbf{M}_d$ 
4:  $T_5 \leftarrow (3d^2 - 2) \cdot R_1$   $\triangleright 1\mathbf{M}_d$ 
5:  $X'_4 \leftarrow -R_1 \cdot (S_1 - T_4) + R_3 \cdot (T_3 + R_4) + R_2 \cdot (T_2 + R_4)$   $\triangleright 3\mathbf{M}$ 
6:  $U'_4 \leftarrow -2S_1 \cdot (T_1 + R_2 + R_3) - R_4 \cdot (T_2 + T_3 + T_5 + 2R_1 + R_4)$   $\triangleright 2\mathbf{M}$ 
7:  $X_4, U_4 \leftarrow X_3 \cdot X'_4, U'_4 X_3 - U_3 X'_4$   $\triangleright 3\mathbf{M}$ 
8: return  $(X_4 : U_4)$   $\triangleright$  Total Cost:  $12\mathbf{M} + 1\mathbf{M}_a + 5\mathbf{M}_d$ 
```

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